



STRATHMORE INSTITUTE OF MATHEMATICAL SCIENCES
MASTER OF SCIENCE IN DATA SCIENCE & ANALYTICS
CAT 2
DSA 8301: Statistical Inference for Big Data

DATE: 20th May 2026

TIME: 1hr 30 Min

Instruction:

Answer All Question

1. Define and explain the following modes of convergence of random variables. Provide the formal definition, notation, and a brief intuition for each case.
 - (a) Convergence in probability (1 marks)
 - (b) Almost sure convergence (1 marks)
 - (c) Convergence in distribution (1 marks)
2. Suppose $X_1, X_2, \dots, X_n$ are independent and identically distributed random variables with:

$$E[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2 < \infty$$

and define the sample mean:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

- (a) State the Weak Law of Large Numbers (WLLN). (2 marks)
- (b) Using Chebyshev's inequality, prove that: (4 marks)

$$\bar{X}_n \xrightarrow{p} \mu$$

- (c) Explain the practical significance of the WLLN in statistical estimation. (2 marks)

3. Let

$$X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2),$$

where σ^2 is known and μ is unknown.

- (a) Write down the probability density function of the normal distribution. **(1 marks)**
- (b) Derive the likelihood function for μ . **(2 marks)**
- (c) Obtain the log-likelihood function. **(1 marks)**
- (d) Derive the score function. **(2 marks)**
- (e) Obtain the maximum likelihood estimator (MLE) of μ . **(2 marks)**
- (f) Verify that the estimator corresponds to a maximum using the second derivative test. **(1 marks)**

4. Two insurance companies wish to compare the average number of claims processed per week.

A random sample from Company A produced: 42, 38, 45, 40, 44, 39, 41, 43

A random sample from Company B produced: 36, 35, 39, 37, 34, 38, 36, 35

Assume the population variances are equal.

At the 5% significance level, test whether the average number of claims processed differs between the two companies.

- (a) State the hypotheses. **(1 marks)**
- (b) Compute the sample means and sample variances. **(3 marks)**
- (c) Compute the pooled variance. **(1 marks)**
- (d) Compute the test statistic. **(2 marks)**
- (e) Determine the rejection region. **(1 marks)**
- (f) State your conclusion. **(2 marks)**